

N. L. Carothers' Real Analysis - Summary of results

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Chapter 1

Compactness

1.1 Uniform Continuity

Definition - Uniform continuity

$f : (M, d) \rightarrow (N, \rho)$ is uniformly continuous if for every $\epsilon > 0$, there exists $\delta > 0$ such that $\rho(f(x), f(y)) < \epsilon$ whenever $d(x, y) < \delta$.

Exercise 8.44

Any Lipschitz map is in fact uniformly continuous. This implies that isometries are also uniformly continuous.

Exercise 8.45

Every map $f : \mathbb{N} \rightarrow \mathbb{R}$ is uniformly continuous. Intuitively, this is because natural numbers never get “arbitrarily close” to each other.

Exercise 48

A uniformly continuous map sends Cauchy sequences to Cauchy sequences.

Exercise 8.54

In bounded, noncompact subsets of $\mathbb{R}$, continuous functions need not be uniformly continuous.

Exercise 8.55

Bounded continuous maps in $\mathbb{R}$ need not be uniformly continuous: for example, $f(x) = \sin(e^x)$, which oscillates “increasingly fast”. Furthermore, unbounded continuous functions *can* be uniformly continuous, as easily seen by the identity function.

Exercise 8.56

A criterion for uniform continuity: $f : (M, d) \rightarrow (N, \rho)$ is uniformly continuous iff $\rho(f(x_n), f(y_n)) \rightarrow 0$ for any pair of sequences $(x_n), (y_n) \subset M$ such that $d(x_n, y_n) \rightarrow 0$.

Exercise 8.57

A type of Lipschitz continuity: If $|f(x) - f(y)| \leq K|x - y|^a$ for $a > 0$ then f is uniformly continuous.

Exercise 8.58

Any function $f : \mathbb{R} \rightarrow \mathbb{R}$ with a bounded derivative is Lipschitz of order 1.

Exercise 8.59

A function satisfying a Lipschitz condition as in Ex. 8.57 for $a > 1$ is **necessarily constant**.

Exercise 8.61

In general, we know that completeness is not necessarily maintained by homeomorphisms. However, if f, f^{-1} are *uniformly continuous* homeomorphisms (**uniform homeomorphisms**) between two metric spaces, then completeness is preserved. We call such metric spaces **uniformly homeomorphic**.

Proposition 8.14

If $f : M \rightarrow N$ is uniformly continuous, then f maps totally bounded sets into totally bounded sets.

Theorem 8.15

If M is a compact metric space, then every continuous map $f : M \rightarrow N$ is uniformly continuous.

Theorem 8.16

Let D be dense in M , let N be complete, and let $f : D \rightarrow N$ be uniformly continuous. Then, f extends uniquely to a uniformly continuous map $F : M \rightarrow N$, defined on all of M . Moreover, if f is an isometry, then so is the extension F .

Corollary 8.17

Completions are unique (up to isometry). That is, if M_1, M_2 are completions of M , then M_1, M_2 are isometric.

1.2 Equivalent Metrics

Proposition 8.19

Suppose that (M, d) is compact and that ρ is another metric on M . Then d, ρ are equivalent iff they are uniformly equivalent.

Theorem 8.20

Let $(V, \|\cdot\|), (V, \|\cdot\|)$ be normed vector spaces, and let $T : V \rightarrow W$ be a linear map. Then the following are equivalent:

- (i) T is Lipschitz
- (ii) T is uniformly continuous
- (iii) T is continuous everywhere
- (iv) T is continuous at $0 \in V$
- (v) there is a constant C such that $\|T(x)\| \leq C\|x\|$ for all $x \in V$

Corollary 8.21

Let $\|\cdot\|_1, \|\cdot\|_2$ be any two norms on a vector space V . Then, $\|\cdot\|_1, \|\cdot\|_2$ are equivalent if and only if there are two constants $0 < c, C < \infty$ such that $c\|x\|_1 \leq \|x\|_2 \leq C\|x\|_1$ for every $x \in V$. In other words, norms are equivalent **iff** they are strongly equivalent.

Theorem 8.22

Any two norms on a finite-dimensional vector space are equivalent.

Corollary 8.23

Let V, W be normed vector spaces with V finite-dimensional. Then, every linear map $T : V \rightarrow W$ is continuous.

Corollary 8.24

Any two finite-dimensional normed vector spaces of the same dimension are uniformly homeomorphic. In fact, we can even find a linear (and hence Lipschitz) homeomorphism between them.

Corollary 8.25

Every finite-dimensional normed vector space is complete.

Corollary 8.26

A finite-dimensional linear subspace of any normed vector space is always closed.

Chapter 2

Category

2.1 Discontinuous Functions

Definition - Oscillation on an interval

Given a bounded interval I and $f : \mathbb{R} \rightarrow \mathbb{R}$, we define $\omega(f; I) = \sup\{|f(x) - f(y)| : x, y \in I\}$. Note: $0 \leq \omega(f; I) \leq 2 \sup_{x \in I} |f(x)|$. If f is unbounded on I , we define $\omega(f; I) = \infty$.

Definition - Oscillation at a point

Given a point $a \in \mathbb{R}$ and $f : \mathbb{R} \rightarrow \mathbb{R}$, we define $\omega_f(a) = \lim_{h \rightarrow 0^+} \text{diam} f(B_h(a))$. If f is unbounded in *every* neighborhood of a , we define $\omega_f(a) = \infty$.

Theorem 9.2

If $f : \mathbb{R} \rightarrow \mathbb{R}$, then $D(f)$ is the **countable** union of **closed** sets in $\mathbb{R}$.